

Database Systems - Normalization Notes

Why normalize? To remove redundancy and the update, insert and delete anomalies.

Functional dependency $X \rightarrow Y$: rows that agree on X must agree on Y .

Candidate key: a minimal set of attributes that determines every attribute.

1NF: every attribute holds a single atomic value; no repeating groups.

2NF: 1NF and no partial dependency of a non-key attribute on part of a composite key.

3NF: 2NF and no transitive dependency (non-key $\rightarrow$ non-key).

BCNF: for every non-trivial dependency $X \rightarrow Y$, X is a superkey.

Exam trap: a table with a single-attribute key is automatically in 2NF.

Exam trap: BCNF is stricter than 3NF; every BCNF table is in 3NF, not the reverse.

Decomposition should be lossless-join and, ideally, dependency-preserving.

Transactions (ACID): Atomicity, Consistency, Isolation, Durability.

Serializability: a concurrent schedule equivalent to some serial schedule.

Two-phase locking (2PL): a growing phase, then a shrinking phase; it guarantees serializability.